

FINANCIAL MANAGEMENT THEORY NOTES FROM MODULE 2 – 6

MODULE -2

Time Value of Money – Theory Notes

1. Meaning of Time Value of Money (TVM)

The **Time Value of Money (TVM)** is a fundamental concept in finance which states that **a sum of money received today is more valuable than the same sum received in the future.**

This is because money available today can be:

- Invested to earn interest or returns.
- Used to generate additional income.
- Affected by inflation if received in the future.
- Subject to risk and uncertainty over time.

Example

₹10,000 received today is more valuable than ₹10,000 received after one year because the ₹10,000 received today can be invested and earn interest.

Importance of TVM

1. Helps in investment decisions.
 2. Helps in capital budgeting.
 3. Used in loan and EMI calculations.
 4. Helps in valuation of securities.
 5. Used in retirement and financial planning.
 6. Helps compare cash flows occurring at different points in time.
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2. Future Value of a Single Cash Flow

Future Value (FV) refers to the value that a present amount of money will grow to after a specified period at a given rate of return or interest.

It is based on the principle of **compounding**.

Formula

$$\begin{aligned} &[\\ &FV = PV(1+r)^n \\ &] \end{aligned}$$

Where:

- **FV** = Future Value
- **PV** = Present Value
- **r** = Rate of interest
- **n** = Number of periods

Factors Affecting Future Value

- Higher present investment results in higher future value.
- Higher interest rate results in higher future value.
- Longer investment period results in higher future value.

Applications

- Investment planning
- Savings decisions
- Retirement planning
- Financial forecasting

3. Future Value of Annuity

An **annuity** is a series of equal cash flows occurring at regular intervals for a specific period.

Examples:

- Monthly savings
- Insurance premiums
- Pension payments
- Regular deposits

The **Future Value of an Annuity** represents the accumulated value of all periodic payments at the end of the investment period.

Formula

$$FV_A = PMT \left[\frac{(1+r)^n - 1}{r} \right]$$

Where:

- **FV_a** = Future Value of Annuity
- **PMT** = Periodic payment
- **r** = Interest rate per period

- **n** = Number of periods

Types of Annuity

1. **Ordinary Annuity** – Payments are made at the end of each period.
2. **Annuity Due** – Payments are made at the beginning of each period.

Applications

- Retirement savings
- Pension funds
- Regular investment plans
- Insurance schemes

4. Present Value and Discounting

Present Value

Present Value (PV) is the current worth of a future amount of money or a series of future cash flows, calculated using a specified discount rate.

The concept is based on the fact that future money is worth less than money available today.

Formula

$$PV = \frac{FV}{(1+r)^n}$$

Factors Affecting Present Value

1. **Higher discount rate → Lower present value**
2. **Longer time period → Lower present value**
3. **Higher future cash flow → Higher present value**

Discounting

Discounting is the process of converting future cash flows into their equivalent value today.

It is the opposite of **compounding**.

Compounding

$$\text{Present Value} \rightarrow \text{Future Value}$$

Discounting

[
Future\ Value \rightarrow Present\ Value
]

Importance of Discounting

- Investment appraisal
 - Capital budgeting
 - Valuation of bonds and shares
 - Loan valuation
 - Project evaluation
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5. Present Value of Single Cash Flow

The **Present Value of a Single Cash Flow** is the current value of one future cash flow.

Formula

[
$$PV = \frac{FV}{(1+r)^n}$$

]

It helps determine how much a future amount is worth today.

Example of Application

If an investor expects to receive ₹1,00,000 after five years, present value calculation helps determine the amount that should be invested today to achieve that future amount.

6. Present Value of Annuity

The **Present Value of Annuity** is the current value of a series of equal payments received or paid at regular intervals.

Formula

[
$$PV_A = PMT \left[\frac{1 - (1+r)^{-n}}{r} \right]$$

]

Applications

- Loan valuation
- EMI calculation

- Lease payments
- Pension valuation
- Insurance payments
- Investment analysis

Types

1. Present Value of Ordinary Annuity
2. Present Value of Annuity Due

7. Present Value of Perpetuity

A **perpetuity** is a series of equal cash flows that continues indefinitely or forever.

The **Present Value of Perpetuity** represents the current value of an infinite series of constant cash flows.

Formula

$$PV = \frac{C}{r}$$

Where:

- **C** = Constant periodic cash flow
- **r** = Discount rate

Characteristics

- Cash flows continue forever.
- Payments are equal and regular.
- There is no fixed maturity period.
- The value depends on the cash flow and discount rate.

Applications

- Valuation of perpetual preference shares
- Endowment funds
- Certain types of financial securities

8. Simple Interest

Simple Interest is the interest calculated only on the original principal amount throughout the entire investment period.

Formula

$$SI = P \times r \times t$$

Where:

- **SI** = Simple Interest
- **P** = Principal
- **r** = Rate of interest
- **t** = Time period

Amount Formula

$$A = P(1 + rt)$$

Characteristics

- Interest is calculated only on principal.
- Interest does not earn further interest.
- Growth is linear.
- Easy to calculate.
- Generally used for short-term financial transactions.

Applications

- Short-term loans
- Certain banking transactions
- Short-term investments

9. Compound Interest

Compound Interest is the interest calculated on the original principal as well as the accumulated interest from previous periods.

It is commonly described as "**interest on interest.**"

Formula

$$A = P(1+r)^n$$

Compound Interest

$$CI = A - P$$

Characteristics

- Interest is added to the principal.
- Future interest is calculated on the increased amount.
- Growth is exponential.
- The longer the period, the greater the effect of compounding.
- Used extensively in banking and investment decisions.

Types of Compounding

1. Annual compounding
2. Semi-annual compounding
3. Quarterly compounding
4. Monthly compounding

10. Difference Between Simple Interest and Compound Interest

Basis	Simple Interest	Compound Interest
Meaning	Interest on original principal	Interest on principal plus accumulated interest
Interest on interest	No	Yes
Growth	Linear	Exponential
Calculation	Easier	More complex
Return over long period	Lower	Higher
Effect of time	Relatively lower	Significant
Common use	Short-term transactions	Long-term investments and loans

11. Capital Recovery Factor (CRF)

The **Capital Recovery Factor (CRF)** is a financial factor used to determine the **equal periodic payment required to recover an initial investment or loan over a specified period at a given interest rate.**

It converts a **present value into an equivalent annual or periodic payment.**

Formula

$$[\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1}]$$

Periodic Payment

$$[\text{PMT} = \text{PV} \times \text{CRF}]$$

Importance

CRF is useful for:

- Loan repayment calculations
- Capital budgeting
- Investment analysis
- Replacement decisions
- Comparing projects with different lives
- Calculating equivalent annual costs

Key Point

CRF converts a present amount into equal periodic payments.

12. Loan Amortization

Loan amortization refers to the systematic repayment of a loan through regular instalments over a specified period.

Each instalment consists of two components:

1. **Interest component**
2. **Principal repayment component**

As the loan is repaid, the outstanding principal decreases.

Loan Amortization Process

At the beginning:

Outstanding Loan → Interest Calculation → EMI Payment → Principal Repayment → Reduced Loan Balance

Components of Amortization

1. Opening Balance

The outstanding loan amount at the beginning of a period.

2. Interest

Interest calculated on the outstanding loan balance.

[
Interest = Opening\ Balance \times Interest\ Rate
]

3. Principal Repayment

The portion of the instalment used to reduce the loan principal.

[
Principal = EMI - Interest
]

4. Closing Balance

The remaining loan balance after principal repayment.

[
Closing\ Balance = Opening\ Balance - Principal\ Repayment
]

13. Loan Amortization Schedule

A **loan amortization schedule** is a table showing the complete repayment of a loan over time.

It normally contains:

Period	Opening Balance	EMI	Interest	Principal	Closing Balance
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Characteristics of Loan Amortization

- Regular payments are made over a fixed period.
- Each payment includes interest and principal.
- Interest is higher during the initial periods.
- Principal repayment increases over time.
- Outstanding loan balance gradually decreases.
- At the end of the loan period, the balance becomes zero.

Applications

- Home loans
 - Vehicle loans
 - Personal loans
 - Education loans
 - Business loans
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14. Important Differences

Compounding vs Discounting

Compounding

Converts present value into future value

Moves cash flow forward in time

Uses future value calculation

Used for investment growth

Discounting

Converts future value into present value

Moves cash flow backward in time

Uses present value calculation

Used for investment valuation

Annuity vs Perpetuity

Annuity

Cash flows continue for a fixed period

Has a specified number of payments

Used in loans, pensions, etc.

Has a maturity period

Perpetuity

Cash flows continue forever

Has an infinite number of payments

Used in perpetual securities and endowments

No maturity period

MODULE – 3

Long-Term Sources of Finance & Cost of Capital – Theory Notes

PART A: LONG-TERM SOURCES OF FINANCE

1. Meaning of Long-Term Finance

Long-term finance refers to funds required by a business for a period generally exceeding **five years**. These funds are mainly used for:

- Purchasing fixed assets
- Expansion and modernization
- Establishing new projects
- Acquisitions and mergers
- Research and development
- Long-term working capital requirements

Long-term sources of finance can be broadly classified into:

1. Owned Funds

- Equity shares
- Preference shares
- Retained earnings

2. Borrowed Funds

- Debentures
- Term loans
- Deferred credit
- Lease financing

3. Alternative Sources

- Venture capital
 - Angel investing
 - Private equity
 - Crowdfunding
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2. Equity Shares

Equity shares represent the ownership capital of a company. Equity shareholders are the real owners of the company and bear the maximum business risk.

Features

- Equity shareholders are owners of the company.
- They have voting rights.
- Dividend is not fixed.
- Dividend is paid only after meeting the claims of preference shareholders and debt holders.
- Equity shareholders have a residual claim on the company's assets.
- Equity capital is a permanent source of finance.
- Equity shareholders bear the highest financial risk.
- They may receive capital appreciation if the company's value increases.

Advantages

- Permanent source of capital.
- No compulsory dividend payment.
- No repayment obligation during the company's lifetime.
- Improves the company's borrowing capacity.
- Useful for financing expansion and growth.

Disadvantages

- Dilution of ownership and control.
 - Equity capital is generally more expensive than debt.
 - Dividend is not tax-deductible.
 - Issue of new shares may reduce existing shareholders' control.
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3. Preference Shares

Preference shares are shares that carry preferential rights over equity shares regarding:

1. Payment of dividend.
2. Repayment of capital during liquidation.

Preference shareholders generally receive a fixed rate of dividend.

Features

- Fixed dividend rate.
- Preference over equity shareholders in dividend payment.
- Preference over equity shareholders in repayment of capital.
- Usually have limited voting rights.
- Can be redeemable or irredeemable, depending on the type.

Types

- Cumulative preference shares
 - Non-cumulative preference shares
 - Participating preference shares
 - Non-participating preference shares
 - Convertible preference shares
 - Non-convertible preference shares
 - Redeemable preference shares
-

4. Debentures

A **debenture** is a long-term debt instrument issued by a company to raise borrowed funds.

Debenture holders are **creditors**, not owners, of the company.

Features

- Fixed rate of interest.
- Interest is generally payable irrespective of profit.
- Debenture holders do not normally have voting rights.
- Debentures have a fixed maturity period.
- They may be secured or unsecured.
- Interest on debt is generally treated as a financial expense.

Advantages

- No dilution of ownership.
- Interest payments are generally tax-deductible.
- Cost of debt is usually lower than cost of equity.
- Suitable for long-term financing.

Disadvantages

- Fixed interest obligation.
- Principal must be repaid at maturity.
- Increases financial risk.
- Failure to pay interest may create financial difficulties.

Types of Debentures

- Secured and unsecured debentures
 - Convertible and non-convertible debentures
 - Redeemable and irredeemable debentures
 - Registered and bearer debentures
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5. Term Loans

A **term loan** is a loan obtained from banks or financial institutions for a specified period, generally used for long-term business requirements.

Features

- Fixed repayment period.
- Regular interest payments.
- Repayment may be made through instalments.
- May be secured against company assets.
- Used mainly for purchasing fixed assets and expansion.

Advantages

- Easy to structure according to business requirements.
- No dilution of ownership.
- Flexible repayment schedules may be available.
- Suitable for financing capital expenditure.

Disadvantages

- Regular interest and principal repayment obligations.
 - Security or collateral may be required.
 - Increases financial risk.
 - May involve restrictive conditions imposed by lenders.
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6. Deferred Credit

Deferred credit is an arrangement where the buyer purchases an asset or equipment but makes payment to the supplier at a future date or in instalments.

It is also known as **supplier's credit** in certain situations.

Features

- Payment is postponed to a future date.
- Useful for purchasing machinery and equipment.
- May involve instalment payments.
- Reduces immediate cash requirements.
- Interest or financing charges may be included.

Advantages

- Reduces immediate financial burden.
- Helps acquire assets without making full upfront payment.
- Convenient for capital-intensive purchases.

Disadvantages

- May involve higher total cost.
 - Supplier may impose credit restrictions.
 - Payment obligations continue even if business performance is weak.
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7. Lease Financing

Lease financing is an arrangement under which the owner of an asset, called the **lessor**, allows another party, called the **lessee**, to use the asset for a specified period in return for periodic lease payments.

Parties

- **Lessor** – Owner of the asset.

- **Lessee** – User of the asset.

Types of Lease

1. Operating Lease

A short-term or cancellable lease where the lessor generally bears ownership-related risks.

2. Finance Lease

A long-term lease where substantially all risks and rewards associated with ownership are transferred to the lessee.

Advantages

- Requires less initial investment.
- Helps conserve working capital.
- Provides access to expensive assets.
- May provide tax benefits depending on applicable laws.
- Flexible financing alternative.

Disadvantages

- Total lease cost may be high.
- Lessee does not initially own the asset.
- Long-term contractual payment obligations.
- Restrictions may be imposed on asset usage.

8. Hybrid Financing

Hybrid financing combines features of both **debt and equity**.

Hybrid instruments provide characteristics of ownership as well as borrowing.

Examples

- Preference shares
- Convertible debentures
- Convertible bonds
- Warrants
- Mezzanine finance

Features

- Combination of debt and equity characteristics.

- May provide fixed returns initially.
- Some instruments can be converted into equity.
- Can help companies raise funds without immediate equity dilution.

Advantages

- Flexible financing option.
- Can reduce immediate ownership dilution.
- May offer tax advantages when debt features are present.
- Useful for companies requiring customized financing.

Disadvantages

- More complex than traditional financing.
- May involve higher costs.
- Conversion may dilute ownership in the future.
- Can create financial obligations.

9. Venture Capital

Venture capital (VC) is a form of equity financing provided to new or growing businesses with high growth potential but also high risk.

Venture capitalists invest funds in exchange for an ownership stake in the business.

Features

- Generally invested in startups and high-growth businesses.
- High risk and high expected return.
- Usually involves equity participation.
- Investors may provide managerial and strategic guidance.
- Investment is generally made with a long-term exit objective.

Advantages

- Provides finance to businesses that may not qualify for bank loans.
- No regular interest payment like traditional debt.
- Provides business expertise and mentoring.
- Helps startups grow rapidly.

Disadvantages

- Founders may lose some ownership and control.
 - Investors may influence important decisions.
 - Venture capitalists expect high returns.
 - Exit pressure may exist.
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10. Angel Investing

Angel investors are wealthy individuals who invest their personal funds in startups and early-stage businesses.

They generally invest in exchange for **equity ownership or convertible securities**.

Features

- Individual investors.
- Focus mainly on startups and early-stage businesses.
- Investment is generally smaller than private equity investments.
- May provide mentoring and business connections.
- Higher risk with potential for high returns.

Advantages

- Easier access to early-stage capital.
- Business guidance and mentoring.
- Access to professional networks.
- Flexible investment arrangements.

Disadvantages

- Ownership dilution.
 - Possible interference in management.
 - Investors may expect significant returns.
 - Relationship conflicts may arise.
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11. Private Equity

Private Equity (PE) refers to investment made by private equity firms or institutional investors in privately held companies or businesses.

Private equity investors generally invest with the objective of improving business performance and earning returns through an eventual exit.

Features

- Large-scale investment.
- Usually focused on established or growing businesses.
- Investors acquire an ownership stake.
- Active involvement in strategic decisions is common.
- Investment is generally for a specified period.

Advantages

- Provides substantial capital.
- Helps business expansion.
- Provides managerial and strategic expertise.
- Can improve corporate governance.
- Helps restructuring and turnaround of businesses.

Disadvantages

- Significant ownership dilution.
- Loss of some managerial control.
- High return expectations.
- Pressure to improve performance and achieve exit.

12. Crowdfunding

Crowdfunding is a method of raising funds from a large number of individuals, usually through an online platform.

Instead of obtaining a large amount from one investor, a business raises smaller amounts from many contributors.

Types of Crowdfunding

1. **Donation-based crowdfunding** – Contributors donate money without expecting financial returns.
2. **Reward-based crowdfunding** – Contributors receive a product, service, or reward.
3. **Equity crowdfunding** – Investors receive ownership shares.
4. **Debt crowdfunding** – Investors lend money and receive interest.

Advantages

- Access to a large number of investors.

- Useful for startups and innovative businesses.
- Helps test market demand.
- Provides marketing and publicity.
- Reduces dependence on traditional finance.

Disadvantages

- Campaigns may fail to reach funding targets.
- Requires significant marketing efforts.
- Risk of disclosure of business ideas.
- Platform fees may apply.
- Regulatory requirements may exist.

PART B: COST OF CAPITAL

13. Meaning of Cost of Capital

Cost of Capital is the minimum rate of return that a company must earn on its investments to maintain the value of the firm and satisfy its investors and financiers.

It represents the **cost incurred by a company for obtaining funds** from various sources.

In simple words:

Cost of capital is the minimum return expected by providers of capital.

Importance

Cost of capital is used for:

- Investment decisions
- Capital budgeting
- Financing decisions
- Capital structure decisions
- Business valuation
- Performance evaluation
- Determining the optimum financing mix

14. Basic Concepts of Cost of Capital

1. Explicit Cost

The actual cost directly associated with raising funds.

2. Implicit Cost

The opportunity cost associated with using funds for a particular purpose.

3. Specific Cost

Cost of obtaining funds from a particular source.

Examples:

- Cost of debt
- Cost of equity
- Cost of preference shares

4. Overall Cost of Capital

The combined cost of all sources of finance.

It is commonly represented by **Weighted Average Cost of Capital (WACC)**.

5. Before-Tax Cost

Cost of financing calculated before considering tax benefits.

6. After-Tax Cost

Cost calculated after considering the tax effect.

15. Components of Cost of Capital

The major components are:

1. **Cost of Debt**
2. **Cost of Debentures**
3. **Cost of Term Loans**
4. **Cost of Preference Capital**
5. **Cost of Equity**
6. **Cost of Retained Earnings**

These individual costs are combined according to their respective weights to determine **WACC**.

16. Cost of Debentures

The **Cost of Debentures** is the effective rate of return required by debenture holders.

Since debentures involve interest payments, the cost of debenture financing is based on the interest obligation and the issue or redemption value.

Important Point

For a company, the **after-tax cost of debt** is generally lower than the before-tax cost because interest is usually tax-deductible.

Factors Affecting Cost of Debentures

- Interest rate
 - Issue price
 - Redemption value
 - Maturity period
 - Tax rate
 - Issue expenses
-

17. Cost of Term Loans

The **Cost of Term Loans** represents the effective cost incurred by a company for obtaining long-term loans from banks or financial institutions.

It primarily includes:

- Interest charges
- Processing fees
- Arrangement fees
- Other borrowing-related costs

The tax deductibility of interest affects the **after-tax cost of term loans**.

Factors Affecting Cost

- Interest rate
 - Loan tenure
 - Security offered
 - Creditworthiness of the company
 - Processing and other charges
 - Tax rate
-

18. Cost of Preference Capital

The **Cost of Preference Capital** is the rate of return expected by preference shareholders.

Since preference shareholders receive a fixed dividend, the cost is related to the dividend payable on preference shares.

Factors Affecting Cost

- Preference dividend rate
- Issue price
- Market price
- Redemption value
- Issue expenses
- Tax considerations

Key Point

Preference dividend is generally **not treated as a tax-deductible expense** because it is an appropriation of profit.

19. Cost of Equity

Cost of Equity is the rate of return expected by equity shareholders for investing in the company's shares.

Equity shareholders take greater risk compared with lenders and preference shareholders. Therefore, the cost of equity is generally higher.

Two major methods for determining the cost of equity are:

1. **Dividend Discount Model (DDM)**
 2. **Capital Asset Pricing Model (CAPM)**
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20. Cost of Equity – Dividend Discount Model (DDM)

The **Dividend Discount Model** determines the cost of equity based on the present value of expected future dividends.

Under the constant-growth model, the cost of equity depends on:

- Expected dividend
- Current market price of equity share
- Expected growth rate of dividends

Basic Concept

The model assumes that the value of an equity share is equal to the present value of its expected future dividends.

Assumptions

- Dividends grow at a constant rate.
- Growth rate remains stable.
- The company is expected to continue its dividend policy.
- Investors require a rate of return based on expected dividends and growth.

Limitations

- Not suitable for companies that do not pay dividends.
 - Constant growth assumption may not be realistic.
 - Difficult to estimate future growth accurately.
 - Market conditions may change.
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21. Capital Asset Pricing Model (CAPM)

The **Capital Asset Pricing Model (CAPM)** determines the expected return on equity based on the risk-free rate, systematic risk, and market risk premium.

Basic Concept

CAPM states that the expected return required by investors depends on:

1. **Risk-free return**
2. **Systematic risk of the security**
3. **Market risk premium**

Components

Risk-Free Rate

Return available from an investment considered to have negligible default risk.

Beta

Measures the systematic risk or sensitivity of a security's return to changes in the overall market.

Market Risk Premium

The additional return investors expect for investing in the market instead of a risk-free investment.

Assumptions of CAPM

- Investors are rational.

- Investors are risk-averse.
- Markets are efficient.
- Investors can diversify their portfolios.
- Systematic risk cannot be eliminated through diversification.
- Investors have similar expectations about risk and return.

Limitations

- Beta may change over time.
 - Risk-free rate may change.
 - Market risk premium is difficult to estimate.
 - Assumptions of perfectly efficient markets may not hold in reality.
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22. Cost of Retained Earnings

Retained earnings are profits retained in the business instead of being distributed as dividends to shareholders.

Although retained earnings do not involve direct cash outflow or external financing costs, they have an **opportunity cost**.

The opportunity cost is the return shareholders could have earned if the profits had been distributed to them and invested elsewhere.

Important Point

Retained earnings are not free funds because shareholders sacrifice alternative investment opportunities when profits are retained.

Factors Affecting Cost of Retained Earnings

- Expected return on alternative investments
 - Shareholders' required rate of return
 - Tax considerations
 - Brokerage and transaction costs
 - Investment opportunities available to shareholders
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23. Weighted Average Cost of Capital (WACC)

Weighted Average Cost of Capital (WACC) is the overall average cost of a company's various sources of finance, weighted according to their proportion in the company's capital structure.

The main sources may include:

- Equity
- Preference shares
- Debt
- Retained earnings

Basic Concept

WACC represents the **average minimum return that a company must earn on its investments to satisfy all providers of capital.**

Importance of WACC

WACC is used for:

- Capital budgeting decisions
- Investment appraisal
- Project evaluation
- Business valuation
- Determining optimal capital structure
- Measuring financial performance
- Discounting future cash flows

Interpretation

- **Project return > WACC** → Generally creates value.
- **Project return = WACC** → Generally maintains value.
- **Project return < WACC** → Generally destroys value.

24. Factors Affecting Cost of Capital

The cost of capital is affected by:

1. **Interest rates**
2. **Business risk**
3. **Financial risk**
4. **Market conditions**
5. **Creditworthiness of the company**
6. **Capital structure**

7. **Tax rates**
 8. **Inflation**
 9. **Economic conditions**
 10. **Investor expectations**
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25. Importance of Cost of Capital

Cost of capital plays an important role in financial management.

1. Investment Decisions

Helps evaluate whether an investment project is financially acceptable.

2. Financing Decisions

Helps select the appropriate source of finance.

3. Capital Structure

Helps determine the suitable combination of debt and equity.

4. Business Valuation

Used as a discount rate in valuation models.

5. Performance Evaluation

Helps assess whether the company is generating sufficient returns.

6. Wealth Maximization

A company aims to earn returns higher than its cost of capital to increase shareholder wealth.

Long-Term Investment Decisions (Capital Budgeting) – Theory Notes

1. Meaning of Capital Budgeting

Capital budgeting is the process of planning, evaluating, and selecting long-term investment projects that involve significant expenditure of funds.

It deals with decisions relating to investments in **long-term assets** such as:

- Land and buildings
- Plant and machinery
- Equipment
- New projects
- Expansion projects
- Replacement of existing assets
- Research and development projects

Capital budgeting decisions are also called **long-term investment decisions**.

Definition

Capital budgeting is the process of evaluating and selecting long-term investment proposals that are expected to generate cash flows over a period of several years.

2. Need for Capital Budgeting

Capital budgeting is needed because long-term investment decisions involve **large amounts of money and long-term consequences**.

The need arises due to:

1. **Large Investment**
 - Capital projects require substantial financial resources.
2. **Long-Term Impact**
 - Decisions affect the company for many years.
3. **Irreversibility**
 - Once investment is made, it may be difficult or costly to reverse.
4. **Risk and Uncertainty**
 - Future cash flows are uncertain.
5. **Limited Resources**
 - Companies have limited funds and must select the best projects.

6. Impact on Profitability

- Good investment decisions increase profitability and shareholder wealth.

7. Growth and Expansion

- Capital budgeting helps businesses plan future expansion.

3. Importance of Capital Budgeting

1. Maximization of Shareholder Wealth

Capital budgeting helps select profitable projects that increase the value of the firm.

2. Long-Term Growth

It helps businesses invest in projects that support future growth and expansion.

3. Efficient Allocation of Funds

Limited financial resources can be allocated to the most profitable projects.

4. Reduction of Risk

Systematic evaluation helps identify and assess risks associated with investment proposals.

5. Better Financial Planning

It helps management estimate future cash inflows and outflows.

6. Replacement Decisions

It helps determine whether old assets should be replaced with new and more efficient assets.

7. Competitiveness

Investment in modern technology and machinery helps companies remain competitive.

8. Strategic Decision-Making

Capital budgeting connects investment decisions with the long-term objectives of the organization.

4. Capital Budgeting Process

The capital budgeting process generally involves the following steps:

Step 1: Identification of Investment Opportunities

The company identifies potential investment opportunities such as:

- New projects
- Expansion
- Modernization

- Replacement
- Diversification

Step 2: Estimation of Investment Requirements

The initial investment required for the project is estimated.

This may include:

- Purchase price of assets
- Installation costs
- Transportation costs
- Working capital requirements

Step 3: Estimation of Cash Flows

Expected future cash inflows and outflows are estimated for the entire life of the project.

Step 4: Evaluation of Investment Proposals

Different capital budgeting techniques are used to evaluate projects.

Examples:

- Payback Period
- Discounted Payback Period
- ARR
- NPV
- IRR
- MIRR
- Profitability Index

Step 5: Selection of Project

The most suitable project is selected based on financial and strategic considerations.

Step 6: Implementation

The selected project is implemented.

Step 7: Monitoring and Review

Actual performance is compared with expected performance to identify deviations.

5. Techniques of Capital Budgeting

Capital budgeting techniques are broadly classified into:

A. Traditional / Non-Discounting Techniques

1. Payback Period
2. Accounting Rate of Return

B. Discounted Cash Flow Techniques

1. Discounted Payback Period
 2. Net Present Value
 3. Internal Rate of Return
 4. Modified Internal Rate of Return
 5. Profitability Index
-

6. Payback Period (PBP)

The **Payback Period** is the time required to recover the initial investment from the project's expected cash inflows.

It measures the **liquidity and risk** of an investment.

Decision Rule

- Shorter payback period → Generally preferred.
- Longer payback period → Generally less preferred.

Advantages

- Simple and easy to understand.
- Easy to calculate.
- Focuses on liquidity.
- Useful for assessing investment risk.
- Suitable for businesses with high uncertainty.

Disadvantages

- Ignores the time value of money.
 - Ignores cash flows after the payback period.
 - Does not directly measure profitability.
 - May result in rejection of profitable long-term projects.
-

7. Discounted Payback Period

The **Discounted Payback Period** is the time required to recover the initial investment using the **present value of future cash flows**.

Unlike the traditional payback period, it considers the **time value of money**.

Decision Rule

A project is generally acceptable if its discounted payback period is within the maximum acceptable period set by the company.

Advantages

- Considers time value of money.
- Considers risk associated with distant cash flows.
- Better than traditional payback period for long-term decisions.

Disadvantages

- Ignores cash flows after the recovery period.
 - More complicated than traditional payback.
 - Does not directly measure total profitability.
-

8. Accounting Rate of Return (ARR)

The **Accounting Rate of Return (ARR)** measures the average accounting profit generated by a project as a percentage of the investment.

It is based on **accounting profit**, rather than cash flows.

Decision Rule

- Higher ARR → Generally preferred.
- Project is accepted if ARR is greater than the predetermined target rate.

Advantages

- Simple to understand.
- Uses accounting information.
- Easy to calculate.
- Considers profits over the entire project life.

Disadvantages

- Ignores time value of money.
- Based on accounting profit rather than cash flows.
- Different accounting methods can affect ARR.

- Does not directly measure shareholder wealth.
-

9. Net Present Value (NPV)

Net Present Value (NPV) is the difference between the present value of future cash inflows and the present value of cash outflows.

It is one of the most important **discounted cash flow techniques**.

Decision Rule

- **NPV > 0** → Accept the project.
- **NPV = 0** → Indifferent.
- **NPV < 0** → Reject the project.

Advantages

- Considers time value of money.
- Considers all cash flows during the project's life.
- Directly relates to shareholder wealth maximization.
- Useful for comparing investment projects.

Disadvantages

- Requires estimation of an appropriate discount rate.
 - More complex than payback period.
 - Results can be affected by inaccurate cash flow estimates.
 - Difficult to compare projects of different sizes using only absolute NPV.
-

10. Internal Rate of Return (IRR)

The **Internal Rate of Return (IRR)** is the discount rate at which the **NPV of a project becomes zero**.

In simple terms, IRR represents the project's expected rate of return based on its projected cash flows.

Decision Rule

- **IRR > Cost of Capital** → Accept.
- **IRR = Cost of Capital** → Indifferent.
- **IRR < Cost of Capital** → Reject.

Advantages

- Considers time value of money.
- Considers all project cash flows.
- Expresses return as a percentage.
- Easy to compare with the cost of capital.

Disadvantages

- May produce multiple IRRs for unconventional cash flows.
 - Can give conflicting results with NPV for mutually exclusive projects.
 - Requires complex calculations.
 - Assumes reinvestment of intermediate cash flows at the IRR.
-

11. Modified Internal Rate of Return (MIRR)

Modified Internal Rate of Return (MIRR) is an improved version of IRR that addresses some limitations of the traditional IRR method.

It assumes that:

- Positive cash flows are reinvested at the firm's required rate of return or cost of capital.
- Negative cash flows are financed at an appropriate financing rate.

Importance

MIRR provides a more realistic measure of project profitability than IRR when the reinvestment assumption of IRR is unrealistic.

Advantages

- Eliminates the problem of multiple IRRs.
- Uses a more realistic reinvestment assumption.
- Considers the time value of money.
- Considers all project cash flows.

Disadvantages

- More complex than NPV.
 - Requires assumptions about reinvestment and financing rates.
 - May still require estimation of appropriate rates.
-

12. Profitability Index (PI)

The **Profitability Index (PI)** is the ratio of the present value of future cash inflows to the initial investment.

It measures the value created per unit of investment.

Decision Rule

- **PI > 1** → Accept.
- **PI = 1** → Indifferent.
- **PI < 1** → Reject.

Advantages

- Considers time value of money.
- Considers all cash flows.
- Useful when capital is limited.
- Helps rank investment projects.

Disadvantages

- May not always give the same ranking as NPV for mutually exclusive projects.
- Requires accurate estimation of cash flows and discount rates.
- Can be misleading when comparing projects of different sizes.

13. Comparison of Capital Budgeting Techniques

Technique	Time Value of Money	Uses Cash Flow	Considers Entire Project Life
Payback Period	No	Yes	No
Discounted Payback	Yes	Yes	No
ARR	No	No, uses accounting profit	Yes
NPV	Yes	Yes	Yes
IRR	Yes	Yes	Yes
MIRR	Yes	Yes	Yes
Profitability Index	Yes	Yes	Yes

Key Point

Among these techniques, **NPV is generally considered the most theoretically sound method** because it directly measures the increase or decrease in the value of the firm.

14. Capital Rationing

Capital rationing refers to a situation where a company has **limited funds available for investment** and cannot undertake all projects with positive returns.

Therefore, the company must select the **best combination of projects** within its available investment budget.

Types

1. Hard Capital Rationing

Occurs when the company faces external limitations on raising additional funds.

2. Soft Capital Rationing

Occurs when management internally imposes a limit on investment spending, even though additional funds may be available.

Importance

- Helps allocate limited funds efficiently.
- Helps select the most profitable combination of projects.
- Prevents excessive investment.
- Helps prioritize investment opportunities.

Role of Profitability Index

Profitability Index is particularly useful in capital rationing because it helps rank projects based on the value generated per unit of investment.

Limitation

The project with the highest PI may not always result in the highest total NPV. Therefore, under capital rationing, management should consider the **overall combination of projects** rather than relying only on PI ranking.

15. Estimation of Cash Flows for New Projects

A **new project** involves investment in a completely new business activity, product, plant, or facility.

The estimation of cash flows is an important step in capital budgeting.

The cash flows generally include:

1. Initial Cash Outflow

The initial investment required at the beginning of the project.

It may include:

- Cost of new asset
- Installation expenses
- Transportation expenses
- Site preparation
- Initial working capital
- Other project-related expenses

2. Operating Cash Flows

These are the annual cash flows generated by the project during its operating life.

They may arise from:

- Sales revenue
- Operating cost savings
- Operating expenses
- Taxes
- Working capital changes

3. Terminal Cash Flow

Cash flow received at the end of the project's life.

It may include:

- Salvage value of assets
- Recovery of working capital
- Tax effects on asset disposal

Important Principles

When estimating project cash flows:

- Only **relevant incremental cash flows** should be considered.
 - **Sunk costs** should generally be ignored.
 - **Opportunity costs** should be included.
 - Changes in working capital should be considered.
 - Tax effects should be considered.
 - Financing costs are generally reflected through the discount rate rather than operating project cash flows.
-

16. Estimation of Cash Flows for Replacement Projects

A **replacement project** involves replacing an existing asset with a new asset.

Examples:

- Replacing old machinery with new machinery.
- Replacing outdated technology.
- Replacing an inefficient vehicle.

The analysis focuses on the **incremental cash flows** resulting from replacing the old asset.

1. Initial Investment

The initial investment generally includes:

- Cost of new asset
- Installation costs
- Less: After-tax proceeds from sale of old asset
- Additional working capital, if required

2. Incremental Operating Cash Flows

The additional cash flow generated by the new asset compared with the old asset.

Benefits may arise from:

- Increased sales
- Lower operating costs
- Lower maintenance costs
- Improved efficiency
- Reduced energy consumption

3. Terminal Cash Flow

At the end of the project's life, cash flows may include:

- Salvage value of new asset
- Recovery of working capital
- Tax effects on disposal

Key Principle

In replacement decisions, only the **difference between the cash flows of the new asset and the old asset** should be considered.

17. New Project vs Replacement Project

Basis	New Project	Replacement Project
Meaning	Investment in a new activity or asset	Replacing an existing asset
Existing Asset	Usually absent	Present
Cash Flow Analysis	Based on total project cash flows	Based on incremental cash flows
Initial Investment	Cost of new project	Cost of new asset less after-tax sale proceeds of old asset
Main Objective	Determine project viability	Determine whether replacement is beneficial

18. Important Capital Budgeting Concepts

Relevant Cash Flows

Cash flows that change as a result of accepting a project.

Incremental Cash Flow

The additional cash flow generated by undertaking a project.

Sunk Cost

A cost that has already been incurred and cannot be recovered. It is generally **ignored** in capital budgeting decisions.

Opportunity Cost

The benefit sacrificed by using a resource for one project instead of its next best alternative. It should be **included** in project cash flows.

Working Capital

Additional investment in current assets required to operate the project. It is generally recovered at the end of the project's life.

Salvage Value

The amount received from selling an asset at the end of its useful life.

Terminal Cash Flow

The final cash flow received at the end of a project's life, including salvage value and recovery of working capital.

MODULE 5

Capital Structure and Dividend Decisions – Theory Notes

PART A: CAPITAL STRUCTURE

1. Meaning of Capital Structure

Capital structure refers to the proportion or combination of different sources of long-term finance used by a company to finance its assets and business activities.

The main components of capital structure are:

- Equity share capital
- Preference share capital
- Retained earnings
- Debentures
- Long-term loans
- Other long-term debt instruments

In simple terms, capital structure shows **how a company finances its long-term operations through debt and equity**.

A company's capital structure can be represented as:

Capital Structure = Debt + Preference Capital + Equity Capital

2. Importance of Capital Structure

Capital structure is important because it directly affects the financial position and profitability of a company.

1. Cost of Capital

The choice of debt and equity affects the overall cost of financing.

2. Financial Risk

Higher debt increases fixed interest obligations and therefore increases financial risk.

3. Return to Shareholders

The use of debt can increase shareholders' returns when the company earns a return greater than the cost of debt.

4. Control of the Company

Equity financing may dilute ownership and voting control, whereas debt generally does not dilute ownership.

5. Financial Flexibility

A suitable capital structure allows the company to raise additional funds when required.

6. Firm Value

An appropriate capital structure can help maximize the value of the firm.

3. Planning the Capital Structure

Capital structure planning is the process of determining the appropriate mix of debt and equity that a company should use to finance its long-term requirements.

The objective is to select a financing mix that balances:

- Cost
- Risk
- Return
- Control
- Financial flexibility

Factors Considered in Capital Structure Planning

1. Business Risk

Companies with high business risk generally prefer lower debt because additional debt increases overall risk.

2. Financial Risk

Debt creates fixed interest and repayment obligations. Excessive debt increases financial risk.

3. Cost of Capital

The company should select sources of finance that minimize its overall cost of capital.

4. Cash Flow Position

Companies with stable and predictable cash flows can generally support higher debt.

5. Control

Issuing additional equity may dilute the ownership and control of existing shareholders.

6. Tax Considerations

Interest on debt is generally tax-deductible, which can make debt financing attractive.

7. Flexibility

The company should maintain the ability to raise funds in the future.

8. Market Conditions

The availability and cost of debt and equity depend on capital market conditions.

9. Size and Age of the Company

Large and established companies generally have easier access to debt and capital markets than small companies.

10. Industry Characteristics

Capital structure requirements differ across industries depending on asset structure, risk, and cash flow patterns.

4. Optimum Capital Structure

Optimum capital structure is the ideal combination of debt and equity that minimizes the company's overall cost of capital and maximizes the value of the firm.

The optimum capital structure aims to achieve a balance between:

Risk and Return

Characteristics

- Minimum overall cost of capital.
- Maximum value of the firm.
- Appropriate balance between debt and equity.
- Acceptable level of financial risk.
- Adequate financial flexibility.
- Efficient use of financial resources.

Objective

The primary objective of optimum capital structure is:

To maximize the market value of the firm and shareholders' wealth.

5. Leverage

Leverage refers to the use of fixed costs in a company's operations or financing structure to magnify the effect of changes in sales or operating profit on shareholders' earnings.

There are three major types:

1. Operating Leverage
2. Financial Leverage
3. Combined Leverage

6. Operating Leverage

Operating leverage arises because of fixed operating costs in the company's cost structure.

Examples of fixed operating costs:

- Rent
- Salaries
- Depreciation
- Insurance

A company with high fixed operating costs has high operating leverage.

Concept

When sales increase, operating profit can increase at a faster rate because fixed operating costs remain unchanged.

Similarly, when sales decrease, operating profit may decline significantly.

Degree of Operating Leverage

$$\left[\text{DOL} = \frac{\text{Contribution}}{\text{EBIT}} \right]$$

Where:

Contribution = Sales – Variable Cost

7. Financial Leverage

Financial leverage arises due to the use of fixed financial charges such as:

- Interest on debt
- Interest on debentures
- Preference dividends

It measures the effect of changes in EBIT on the earnings available to equity shareholders.

Degree of Financial Leverage

$$\left[\text{DFL} = \frac{\text{EBIT}}{\text{EBT}} \right]$$

Where:

- **EBIT** = Earnings Before Interest and Tax
- **EBT** = Earnings Before Tax

Concept

If a company uses debt and earns a return greater than the cost of debt, financial leverage can increase the return available to equity shareholders.

However, excessive debt increases financial risk.

8. Combined Leverage

Combined leverage represents the combined effect of operating leverage and financial leverage.

It shows how changes in sales can affect the earnings available to equity shareholders.

Formula

$$[\text{DCL} = \text{DOL} \times \text{DFL}]$$

or

$$[\text{DCL} = \frac{\text{Contribution}}{\text{EBT}}]$$

A company with high combined leverage has greater overall risk because both operating and financial fixed costs are high.

9. Governance of Equity and Debt Leverages

The management of equity and debt leverage involves maintaining an appropriate balance between borrowed funds and shareholders' funds.

Equity Financing

Equity does not create compulsory interest payments but may dilute ownership and control.

Debt Financing

Debt provides financial leverage but creates fixed obligations such as:

- Interest payments
- Principal repayment

Effective Leverage Management

A company should:

- Maintain an appropriate debt-equity ratio.
- Ensure sufficient cash flows to service debt.
- Avoid excessive financial risk.
- Use debt when the expected return exceeds its cost.
- Maintain sufficient borrowing capacity.
- Consider business risk and financial risk together.

The objective is to achieve **maximum benefit from leverage while maintaining acceptable risk.**

10. EBIT-EPS Analysis

EBIT-EPS analysis is a technique used to examine the effect of different financing plans on the earnings available to equity shareholders.

It helps management choose between alternative financing plans involving:

- Equity shares
- Debt
- Preference shares

EBIT

EBIT means **Earnings Before Interest and Tax.**

EPS

EPS means **Earnings Per Share.**

It represents the earnings available to each equity share.

Purpose of EBIT-EPS Analysis

- Compare alternative financing plans.
- Determine the effect of financial leverage.
- Identify the financing plan that provides higher EPS.
- Assist in capital structure decisions.
- Determine the level of EBIT at which two financing plans produce the same EPS.

Indifference Point

The **EBIT-EPS indifference point** is the level of EBIT at which the EPS under two alternative financing plans is equal.

Advantages

- Easy to understand.
- Helps compare financing alternatives.
- Shows the effect of debt financing on EPS.
- Useful in capital structure planning.

Limitations

- Focuses mainly on EPS and may ignore risk.
 - Ignores market value of shares.
 - Does not consider qualitative factors.
 - Higher EPS does not always mean higher shareholder wealth.
-

11. Return on Investment (ROI)

Return on Investment (ROI) measures the profitability or return generated by an investment relative to the amount invested.

It is used to evaluate the efficiency of an investment.

Concept

A higher ROI indicates that the investment is generating a higher return relative to the funds invested.

Importance

- Measures investment efficiency.
 - Helps compare different investment opportunities.
 - Used in performance evaluation.
 - Helps management assess profitability.
-

12. Return on Equity (ROE)

Return on Equity (ROE) measures the return generated by a company on the shareholders' equity invested in the business.

It indicates how efficiently the company uses shareholders' funds to generate profits.

Concept

A higher ROE generally indicates better utilization of shareholders' capital.

Importance

- Measures shareholder profitability.

- Helps assess management efficiency.
- Useful for comparing companies.
- Indicates the earning capacity of shareholders' investment.

ROI vs ROE

Basis	ROI	ROE
Meaning	Return generated on investment	Return generated on shareholders' equity
Focus	Overall investment efficiency	Equity shareholders' return
Users	Management and investors	Mainly equity shareholders and investors
Purpose	Evaluate investment performance	Evaluate return to owners

PART B: DIVIDEND DECISIONS

13. Meaning of Dividend

A **dividend** is the portion of a company's profit distributed to its shareholders.

The company may either:

- Distribute profits as dividends, or
- Retain profits for future business requirements.

Therefore, dividend decisions involve determining:

How much profit should be distributed to shareholders and how much should be retained in the business.

14. Dividend Decision

A **dividend decision** is a financial decision concerning the distribution of a company's profits among shareholders.

The management must decide:

1. Whether to pay dividends.
2. How much dividend to pay.
3. How frequently to pay dividends.
4. How much profit to retain for future investment.

The dividend decision is closely related to the company's:

- Investment decisions

- Financing decisions
 - Growth plans
-

15. Dividend Policy

Dividend policy refers to the company's approach or strategy regarding the distribution of profits to shareholders.

It determines the proportion of earnings paid as dividends and the proportion retained in the business.

Objectives of Dividend Policy

- Provide income to shareholders.
 - Maintain shareholder confidence.
 - Support the company's growth.
 - Maintain financial stability.
 - Maximize shareholder wealth.
-

16. Factors Affecting Dividend Policy

1. Profitability

Companies with higher and stable profits generally have greater ability to pay dividends.

2. Stability of Earnings

Companies with stable earnings can maintain regular dividend payments.

3. Cash Flow Position

Dividend payments require sufficient cash availability.

4. Growth Opportunities

Companies with significant investment opportunities may retain more profits.

5. Access to Capital Markets

Companies with easy access to external finance may distribute more profits as dividends.

6. Shareholders' Expectations

Investors may prefer regular and stable dividend income.

7. Legal Restrictions

Companies must comply with applicable legal and regulatory requirements regarding dividend distribution.

8. Tax Considerations

Tax treatment of dividends and capital gains can influence investor preferences and company policy.

9. Debt Obligations

Companies with significant debt commitments may retain profits to meet interest and repayment obligations.

10. Liquidity

A profitable company may still have limited cash available for dividend payments.

11. Control Considerations

Retaining profits can reduce the need to issue new shares, thereby helping existing shareholders maintain control.

12. Inflation

During inflation, companies may need more funds to replace assets and maintain operations, leading to higher retention.

17. Types of Dividend Policy

1. Stable Dividend Policy

Under this policy, the company pays a stable and consistent dividend over time.

Features

- Stable dividend per share.
- Provides certainty to shareholders.
- Builds investor confidence.

Advantage

Suitable for shareholders who prefer regular income.

2. Constant Payout Ratio Policy

Under this policy, the company pays a fixed percentage of earnings as dividends.

For example, if the payout ratio is fixed, the dividend changes according to changes in profits.

Features

- Dividend varies with earnings.
 - Maintains a fixed relationship between profit and dividend.
-

3. Regular Dividend Plus Extra Dividend Policy

Under this policy, the company pays a regular dividend and declares an additional dividend when profits are exceptionally high.

Features

- Provides a stable basic dividend.
 - Extra dividend is paid during highly profitable periods.
 - Gives flexibility to the company.
-

4. Residual Dividend Policy

Under this policy, the company first finances all profitable investment opportunities using retained earnings and distributes the remaining profits as dividends.

Process

Profits → Investment Requirements → Retained Earnings → Remaining Amount Distributed as Dividend

18. Forms of Dividend

Dividends can be distributed in different forms.

1. Cash Dividend

Dividend paid directly in cash to shareholders.

It is the most common form of dividend.

2. Stock Dividend

Dividend distributed in the form of additional shares instead of cash.

It is commonly referred to as a **bonus issue** when issued from eligible reserves according to applicable regulations.

3. Property Dividend

Dividend paid in the form of assets or property other than cash.

4. Scrip Dividend

Dividend paid through a written promise or document that the company will pay the dividend at a later date.

5. Bond Dividend

Dividend paid through bonds or debt instruments issued by the company.

19. Bonus Issue

A **bonus issue** refers to the issue of additional fully paid shares to existing shareholders without requiring them to make additional payment.

Bonus shares are generally issued by capitalizing eligible reserves.

Features

- No cash payment is required from shareholders.
- Existing shareholders receive additional shares.
- Increases the number of shares outstanding.
- Does not directly bring new funds into the company.
- May improve share liquidity.
- Shareholders' proportional ownership generally remains unchanged.

Objectives

- Reward existing shareholders.
- Increase the number of shares in circulation.
- Improve marketability of shares.
- Signal management's confidence in the company.
- Convert accumulated reserves into share capital.

Effect

A bonus issue does not directly increase the total net worth of the company because reserves are converted into share capital.

20. Stock Split

A **stock split** is the division of existing shares into a larger number of shares with a lower face value per share.

For example, in a **1:2 stock split**, one existing share may be divided into two shares, subject to the applicable terms of the split.

Features

- Number of shares increases.
- Face value per share decreases proportionately.
- Total share capital remains unchanged.

- No new funds are raised.
- Existing shareholders' ownership proportion remains unchanged.
- May improve share liquidity and affordability.

Objectives

- Make shares more affordable to investors.
- Improve liquidity.
- Increase market participation.
- Broaden the investor base.

21. Bonus Issue vs Stock Split

Basis	Bonus Issue	Stock Split
Meaning	Free additional shares issued to existing shareholders	Existing shares divided into smaller denomination shares
Source	Capitalization of eligible reserves	Division of existing share capital
Face Value	Usually remains the same	Decreases
Number of Shares	Increases	Increases
Total Share Capital	Generally unchanged	Unchanged
Cash Inflow	No	No
Ownership Proportion	Unchanged	Unchanged
Main Objective	Capitalize reserves and reward shareholders	Improve affordability and liquidity

22. Relationship Between Capital Structure and Dividend Decisions

Capital structure and dividend decisions are interconnected financial decisions.

- **Capital structure decision** determines how the company finances its assets.
- **Dividend decision** determines how profits are distributed or retained.

If a company pays a high dividend:

- Less profit is retained.

- The company may need more external financing.
- Debt or equity financing may increase.

If a company retains more profit:

- Internal financing increases.
- Dependence on external financing may decrease.
- The company may have more funds for expansion.

Thus, the company must balance:

Dividend Distribution + Retained Earnings + External Financing

The overall objective is to achieve **financial stability and maximize shareholders' wealth**.

MODULE - 6

Working Capital Management – Theory Notes

1. Meaning of Working Capital

Working Capital refers to the funds invested in the **current assets** of a business to meet its day-to-day operating requirements.

Current assets generally include:

- Cash and bank balances
- Marketable securities
- Inventory or stock
- Trade receivables or debtors
- Bills receivable
- Short-term advances

Working capital is essential for the smooth functioning of the business.

2. Types of Working Capital

A. Gross Working Capital

Gross working capital refers to the **total investment in current assets**.

$$\begin{aligned} &[\\ \text{Gross \ Working \ Capital} &= \text{Total \ Current \ Assets} \\ &] \end{aligned}$$

B. Net Working Capital

Net working capital is the excess of current assets over current liabilities.

$$\begin{aligned} &[\\ \text{Net \ Working \ Capital} &= \text{Current \ Assets} - \text{Current \ Liabilities} \\ &] \end{aligned}$$

C. Permanent Working Capital

Permanent working capital is the **minimum level of current assets that a business must maintain continuously** to carry out its normal operations.

D. Temporary Working Capital

Temporary working capital is the additional working capital required to meet **seasonal or temporary increases in business activity**.

3. Working Capital Management

Working Capital Management refers to the planning and control of current assets and current liabilities to ensure that the business has sufficient liquidity while maintaining profitability.

The main objective is to maintain a proper balance between:

Liquidity and Profitability

Too much working capital may reduce profitability because funds remain idle.

Too little working capital may create liquidity problems and affect the firm's ability to meet short-term obligations.

4. Sources of Working Capital

Working capital can be financed through **short-term and long-term sources**.

A. Short-Term Sources

1. Trade Credit

Trade credit is credit provided by suppliers when goods are purchased on credit.

It is one of the most common sources of short-term finance.

2. Bank Credit

Banks provide short-term finance through:

- Cash credit
- Overdraft
- Short-term loans
- Bills discounting

3. Commercial Paper

Commercial paper is an unsecured short-term money market instrument issued by financially strong companies to raise short-term funds.

4. Accrued Expenses

These are expenses that have been incurred but not yet paid.

Examples:

- Salaries payable
- Wages payable
- Taxes payable

5. Customer Advances

Businesses may receive advance payments from customers before supplying goods or services.

B. Long-Term Sources

1. Equity Share Capital

Funds raised by issuing equity shares.

2. Preference Share Capital

Funds raised through preference shares.

3. Retained Earnings

Profits retained in the business can be used to finance working capital.

4. Long-Term Loans

Banks and financial institutions may provide long-term funds that can support permanent working capital requirements.

5. Debentures

Long-term debt instruments issued by companies.

5. Factors Influencing Working Capital Requirements

The amount of working capital required varies from business to business.

1. Nature of Business

Manufacturing companies generally require more working capital than service businesses.

2. Size of Business

Large businesses usually require greater working capital in absolute terms.

3. Length of Operating Cycle

A longer operating cycle requires greater investment in working capital.

4. Production Cycle

Businesses with long production processes require more funds to maintain inventory and production activities.

5. Credit Policy

A liberal credit policy increases investment in trade receivables.

A strict credit policy reduces the requirement for working capital.

6. Inventory Management

Higher inventory levels increase working capital requirements.

Efficient inventory management reduces the amount of funds blocked in stock.

7. Growth and Expansion

Growing businesses generally require more working capital to support increased sales and production.

8. Seasonal Fluctuations

Seasonal businesses require additional working capital during peak seasons.

9. Business Cycle

During periods of economic expansion, working capital requirements may increase due to higher sales.

10. Availability of Credit

Easy access to bank finance and trade credit can reduce the need for internally maintained working capital.

11. Price Level Changes

Inflation increases the cost of raw materials, wages, and other expenses, thereby increasing working capital requirements.

12. Operating Efficiency

Efficient production, inventory, collection, and payment systems reduce working capital requirements.

13. Payment Policy

A business that receives longer credit from suppliers may require less working capital.

6. Current Asset Policy

Current asset policy refers to the policy adopted by a firm regarding the level of current assets it maintains in relation to its sales and business operations.

There are three major current asset investment policies.

A. Conservative Current Asset Policy

Under a conservative policy, the firm maintains a **high level of current assets**.

Features

- High liquidity
- Low risk
- Lower probability of stock-outs
- Higher investment in current assets
- Lower profitability due to idle funds

Suitable For

Businesses that prioritize safety and liquidity.

B. Aggressive Current Asset Policy

Under an aggressive policy, the firm maintains a **low level of current assets**.

Features

- Lower investment in current assets
- Higher risk
- Higher potential profitability
- Greater possibility of liquidity problems

Suitable For

Businesses willing to accept higher risk for potentially higher returns.

C. Moderate or Matching Policy

Under this approach, the firm maintains a balance between liquidity and profitability.

The level of current assets is maintained at a reasonable level based on operating requirements.

7. Current Asset Financing Policy

Current asset financing policy refers to the method used by a company to finance its current assets.

The major policies are:

A. Conservative Financing Policy

Under this policy, the company uses **long-term funds to finance permanent working capital and a portion of temporary working capital**.

Features

- Lower financial risk
 - Higher liquidity
 - Higher financing cost
 - Greater financial safety
-

B. Aggressive Financing Policy

Under this policy, the company uses **short-term funds to finance a significant portion of permanent working capital**.

Features

- Lower financing cost
 - Higher financial risk
 - Greater refinancing risk
 - Lower liquidity
-

C. Matching or Hedging Policy

Under this policy, the maturity of financing is matched with the life of the assets.

- Permanent assets → Long-term finance
- Temporary assets → Short-term finance

The objective is to match the duration of assets and liabilities.

8. Operating Cycle

The **Operating Cycle** is the period between the acquisition of raw materials and the ultimate collection of cash from the sale of finished goods.

It represents the time taken by a business to convert:

Cash → Raw Materials → Work-in-Progress → Finished Goods → Sales → Receivables → Cash

Components of Operating Cycle

1. **Raw Material Storage Period**
2. **Work-in-Progress Conversion Period**
3. **Finished Goods Storage Period**
4. **Receivables Collection Period**

Formula

[
Operating\ Cycle =
RM\ Storage\ Period

- WIP\ Period
 - Finished\ Goods\ Period
 - Receivables\ Period
-]

The operating cycle indicates the period for which funds remain tied up in the operating process.

Importance

- Helps estimate working capital requirements.
 - Helps identify delays in operations.
 - Helps improve operating efficiency.
 - Helps determine liquidity requirements.
-

9. Cash Cycle

The **Cash Cycle**, also called the **Cash Conversion Cycle**, measures the time between the payment of cash to suppliers and the collection of cash from customers.

It represents the period for which the firm's own funds remain tied up in the operating cycle.

Formula

$$\begin{aligned} & [\\ \text{Cash Cycle} &= \text{Operating Cycle} - \text{Payables Deferral Period} \\ &] \end{aligned}$$

The **Payables Deferral Period** is the average time taken by the company to pay its suppliers.

Cash Cycle Process

Payment to Suppliers → Production → Sale → Collection from Customers

A shorter cash cycle generally means that the company can recover its investment in working capital more quickly.

10. Difference Between Operating Cycle and Cash Cycle

Basis	Operating Cycle	Cash Cycle
Meaning	Time from purchase of raw materials to collection from customers	Time between payment to suppliers and collection from customers
Supplier Credit	Not deducted	Deducted
Focus	Entire operating process	Actual period funds are tied up
Formula	Inventory period + Receivables period	Operating cycle – Payables period
Purpose	Measures operating efficiency	Measures working capital financing requirement

11. Estimation of Working Capital Requirements

Estimation of working capital requirements means determining the amount of funds needed to finance the firm's day-to-day operations.

The objective is to ensure that sufficient funds are available to meet operating expenses without maintaining excessive idle funds.

The estimation generally involves calculating:

A. Current Assets

1. Raw material inventory
2. Work-in-progress
3. Finished goods
4. Trade receivables
5. Cash and bank balance
6. Other current assets

B. Current Liabilities

1. Trade creditors
2. Bills payable
3. Outstanding expenses
4. Other short-term liabilities

Net Working Capital Requirement

[
Net\ Working\ Capital = Total\ Current\ Assets - Total\ Current\ Liabilities
]

12. Steps in Estimating Working Capital Requirements

Step 1: Estimate Sales and Production

Forecast the expected level of sales and production.

Step 2: Estimate Operating Expenses

Determine the costs related to:

- Raw materials
- Labour
- Manufacturing

- Administration
- Selling and distribution

Step 3: Estimate Current Assets

Calculate the required investment in:

- Raw materials
- Work-in-progress
- Finished goods
- Receivables
- Cash

Step 4: Estimate Current Liabilities

Determine expected financing from:

- Trade creditors
- Bills payable
- Outstanding expenses
- Other current liabilities

Step 5: Determine Net Working Capital

Subtract current liabilities from current assets.

Step 6: Add Safety Margin

An additional margin may be maintained to deal with uncertainties and unexpected changes in business conditions.

13. Factors to Consider While Estimating Working Capital

While estimating working capital requirements, the firm should consider:

- Expected sales volume
- Production volume
- Operating cycle
- Inventory holding period
- Receivables collection period
- Credit period received from suppliers
- Credit policy offered to customers

- Seasonal fluctuations
 - Growth rate
 - Business conditions
 - Price level changes
 - Operating efficiency
 - Availability of short-term finance
-

14. Importance of Working Capital Estimation

Proper estimation helps a firm to:

1. Maintain adequate liquidity.
 2. Avoid shortage of funds.
 3. Avoid excessive investment in current assets.
 4. Reduce financing costs.
 5. Maintain smooth production.
 6. Pay suppliers and employees on time.
 7. Improve profitability.
 8. Manage seasonal fluctuations.
 9. Maintain good creditworthiness.
 10. Ensure uninterrupted business operations.
-

15. Liquidity vs Profitability in Working Capital Management

Working capital management involves a trade-off between **liquidity and profitability**.

Higher Working Capital

- Higher liquidity
- Lower risk
- Lower profitability

Lower Working Capital

- Lower liquidity
- Higher risk
- Higher potential profitability

Therefore, the objective of working capital management is to find an **optimum level of working capital** that balances liquidity, risk, and profitability.

16. Summary

Working Capital is the investment in current assets required for day-to-day operations.

Sources of Working Capital include trade credit, bank finance, commercial paper, customer advances, equity, retained earnings, and long-term borrowings.

Working Capital Requirements are influenced by business nature, size, operating cycle, production cycle, credit policy, inventory, growth, seasonality, inflation, and operating efficiency.

Current Asset Policy determines the level of current assets maintained by a firm and may be conservative, aggressive, or moderate.

Current Asset Financing Policy determines how current assets are financed and may be conservative, aggressive, or matching.

Operating Cycle measures the time from acquisition of raw materials to collection of cash from customers.

Cash Cycle measures the time between payment to suppliers and collection from customers.

Working Capital Estimation involves estimating current assets and current liabilities and determining the net working capital required to operate the business smoothly.